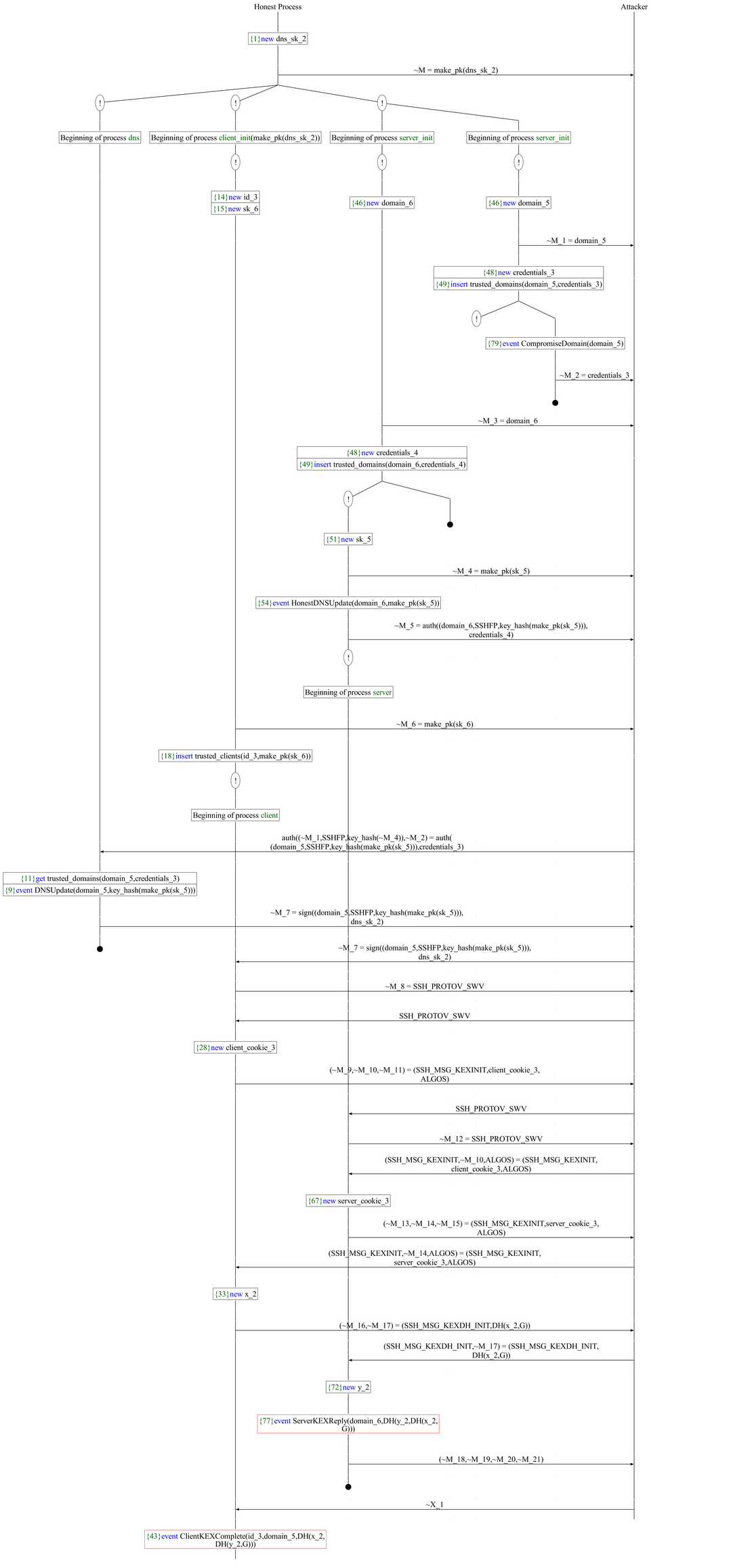
Abbreviations
~M_18 = SSH_MSG_KEXDH_REPLY
~M_19 = make_pk(sk_5)
~M_20 = DH(y_2,G)
~M_21 = sign(hash((SSH_PROTOV_SWV,SSH_PROTOV_SWV,(SSH_MSG_KEXINIT,client_cookie_3,ALGOS),(SSH_MSG_KEXINIT,server_cookie_3,ALGOS),make_pk(sk_5),DH(x_2,G),DH(y_2,G),DH(y_2,DH(x_2,G)))),sk_5)
~X_1 = (SSH_MSG_KEXDH_REPLY,(~M_4,~M_20,~M_21)) = (SSH_MSG_KEXDH_REPLY,(make_pk(sk_5),DH(y_2,G),sign(hash((SSH_PROTOV_SWV,SSH_PROTOV_SWV,(SSH_MSG_KEXINIT,client_cookie_3,ALGOS),(SSH_MSG_KEXINIT,server_cookie_3,ALGOS),make_pk(sk_5),DH(x_2,G),DH(y_2,G),DH(y_2,DH(x_2,G)))),sk_5)))

A trace has been found.